# MD-DGFLS: MULTI-DOMAIN DEEPFAKE FORENSIC VERIFICATION CERTIFICATE

IEEE TPAMI / CVPR Technical Forensics Specification v2.2-PROD | Cryptographic Audit Dossier

| | | | |
|---|---|---|---|
| **File Name:** | audit.png | **Target Domain:** | Face |
| **Audit Timestamp:** | 2026-10-05 09:14:45 UTC | **SHA-256 Hash:** | 99a0698a9697e6c71465ddd8... |
| **Forensic Verdict:** | **VERIFIED AUTHENTIC MEDIA** | **Confidence:** | **68.9%** |
| **Tamper Type:** | Authentic Media Stream | **Bounding Box:** | None |

## Three-Stage Visual Forensics Pipeline



**Stage 1: Source Image**




**Stage 2: Tamper Overlay**



**Stage 3: XAI Heatmap**

## Mathematical Innovations & Multi-Expert Telemetry

| | |
|---|---|
| **CDFAM (Cross-Domain Alignment)** | Contrastive Score: 0.4778 (Margin: m=0.3) |
| **ARCC (Entropy Calibration)** | Dynamic Calibration Temperature: T = 1.628 |
| **TGHFL (Transformer-GAN Cross-Attention)** | Cross-Attention Energy: 0.0138 |
| **MEFCN (Multi-Expert Consensus)** | Face: 76.0% \| Medical: 4.9% \| Satellite: 7.2% \| Regular: 6.0% \| Natural: 6.0% |
| **HLRN (Localization Refinement)** | Coarse-to-Fine Refinement: 556x660 Native |
| **Frequency Stream (ELA & Laplacian)** | ELA Discrepancy: 0.74 \| Sensor Noise Var: 313.9 |

This certificate is cryptographically anchored and generated by the MD-DGFLS Autonomous Forensics Engine. Verified under IEEE TPAMI standards.